

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		award (1) each for A and B A copper(II) nitrate / copper nitrate / Cu(NO ₃) ₂ B carbon dioxide / CO ₂ award (1) for both C and D C zinc nitrate / Zn(NO ₃) ₂ D copper	3			3		3
		(ii)		Mg + 2HNO ₃ → Mg(NO ₃) ₂ + H ₂ award (1) each for formulae of <u>products</u> award (1) for balancing only if all formulae are correct		3		3		
		(iii)		displacement	1			1		1
	(b)	(i)		$\text{H}^+ + \text{Cl}^- \longrightarrow \text{HCl}$ ☐ $\text{Ba}^{2+} + \text{SO}_4^{2-} \longrightarrow \text{BaSO}_4$ ☒ $\text{Ba}^{2+} + \text{S}^{2-} + 4\text{O}^{2-} \longrightarrow \text{BaSO}_4$ ☐ $2\text{H}^+ + 2\text{Cl}^- \longrightarrow 2\text{HCl}$ ☐ $\text{Ba}^{2+} + \text{SO}_4^{2-} \longrightarrow \text{BaSO}_4$ ☐			1	1		

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		(ii)	I	flame test – (apple) green (flame)	1			1		1
			II	silver nitrate (solution) – white precipitate	1			1		1
				Question 5 total	6	3	1	10	0	6